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MILITARY SPECIFICATION

PACKAGING, HANDLING, AND TRANSPORTABILITY IN SYSTEM/EQUIPMENT ACQUISITION

1. SCOPE

1.1 Purpose. This specification establishes packaging, materials handling, and transportability requirements and criteria for acquisition of systems and equipment. It requires an interface between these functions and the system engineering management process as set forth in MIL-STD-499 and related configuration management requirements contained in the applicable MIL-STD-480 series directives and MIL-STD-490.

1.1.1 Acquisition. Acquisition encompasses the life cycle phases for a given system/equipment program. The life cycle will generally (but not necessarily) consist of five phases: Conceptual, validation, full-scale development, production, and deployment. A program, dependent upon its scope, may skip a phase or have program elements in any or all other phases.

1.2 Applicability. The requirements of this specification may be contractually applied in total, or tailored to meet the specific needs of a system, equipment, or item acquisition. Tailoring as used in this specification means selective adaptation of the requirements of this specification through deletion or alteration of the specific characteristics and requirements of the system, equipment, or item being acquired.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-B-131	Barrier Material, Water Vaporproof, Flexible
MIL-D-1000	Drawings, Engineering and Associated Lists

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MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-S-19491	Semiconductor Device, Packaging of
MIL-F-22191	Films, Transparent, Flexible, Heat Sealable, for Packaging Applications
MIL-A-25175	Air Transport, Nontactical, Packing for
MIL-I-26860	Indicator, Humidity, Plug, Color Change
MIL-C-26861	Cushioning Material, Resilient Type, General
MIL-V-27166	Valve, Pressure Equalizing, Gaseous Products
MIL-P-27443	Pallet, Cargo, Aircraft, Type HCU-6/E, HCU-12/E, and HCU-10/C
MIL-N-27444	Net, Cargo Tiedown, Pallets HCU-7/E, HCU-15/C, HCU-11/C, and HCU-16/C
MIL-V-38352	Value Engineering Program Requirements
MIL-C-45662	Calibration System Requirements
MIL-M-45907	Mount, Shipping Container, Resilient, Shock and Vibration Damping
MIL-M-55565	Microcircuits, Packaging of
MIL-P-81705	Plastic Sheets, Flexible, Electrostatic-Free, Heat Sealable, Heavy Duty, for Packaging Applications
MIL-R-83313	Reviews and Audits, Technical, for Communication-Electronic-Meteorological Systems and Related Equipment
MIL-P-84000	Provisioning Screening Data to be Furnished by Government Suppliers

STANDARDS

Federal

FED-STD-101	Preservation, Packaging, and Packing Materials, Test Procedures

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-209	Slinging Eyes and Attachments for Lifting and Tying Down Heavy Military Equipment
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers
MIL-STD-481	Configuration Control - Engineering Changes, Deviations and Waivers (Short Form)
MIL-STD-483	Configuration Management Practices for Systems, Equipment, Munitions, and Computer Programs
MIL-STD-490	Specification Practices
MIL-STD-499	System Engineering Management

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MIL-STD-669	Loading Environment and Related Requirements for Platform Rigged Airdrop Materiel
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-814	Requirements for Tiedown, Suspension and Extraction Provisions on Military Materiel for Airdrop
MIL-STD-831	Test Reports, Preparation of
MIL-STD-834	Packaging Data Forms, Instructions for Preparation and Use of
MIL-STD-881	Work Breakdown Structures for Defense Materiel Items
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods
MS20003	Indicator, Humidity, Card, Three Spot, Impregnated Areas (Cobaltous Chloride)

PUBLICATIONS

Military Handbook

MIL-HDBK-304	Package Cushioning Design

Air Force Bulletin

518	Cargo Aircraft Compartment Dimensional Data

Regulations

Department of Defense

4500.32R	Military Standard Transportation and Movement Procedures

Armed Services Procurement

7-104.79	Safety Precautions for Ammunition and Explosives
1-1701.1	Value Engineering Incentives

Air Force

Manual

71-4	Packaging and Handling of Dangerous Materials for Transportation by Military Aircraft

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Association of American Railroads

The Official Railway Equipment Register of the United States

Car Builders Encyclopedia

(Application for copies should be addressed to the American Railway Engineering Association, 59 East Van Buren Street, Chicago, Illinois 60605.)

National Aerospace

Standards

NAS 850 General Packaging Standard

(Applications for copies should be addressed to the Aerospace Industries Association of America, Incorporated, 1725 DeSales Street, Washington, D. C. 20036.)

Department of Transportation

Title 14, 46, 49 Code of Federal Regulations

(Applications for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D. C. 20591.)

3. REQUIREMENTS

3.1 Program management requirements. An early and orderly identification and resolution of transportation, packaging, handling, and transportability requirements and constraints shall be accomplished to coincide with contractually imposed program events and milestones (see 3.1.1). This includes insuring that appropriate requirements of this specification are contractually required of and adequately performed by lower-tier contractors for shipments which are made direct to government activities. Packaging, handling, transportability, and transportation requirements and constraints shall be interfaced with and, insofar as practical, integrated into the design of program hardware to achieve the lowest program life cycle cost. (See 3.2 for definition of transportability.)

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a. Transportability of items and equipment, and technical adequacy of packaging designs shall be consistent with program test, deployment, employment and logistics support concepts and plans.

b. Handling and mobility features shall be incorporated into items, equipment, and containers as required to facilitate handling and movement consistent with existing equipment, facilities, and procedures.

c. Design, test, and production of transportation and handling equipment, containers, and packaging needed to conduct and support development and operational test and evaluation (DT&E and OT&E) and production shall be so accomplished as to assure their availability at the time and place required, and shall be suitable for use throughout the system program without further design and test effort unless program objectives, cost, or schedules justify otherwise, or if deficiencies are detected which require correction (see item f).

d. The most cost-effective item/container design interface shall be determined utilizing trade study techniques based upon program life cycle costs. Trade studies and transportability analyses required by MIL-STD-499, or as otherwise required, shall be conducted to insure compatibility between item, transportation, and handling equipment and to determine the most efficient and cost effective packaging design required to satisfy functional requirements (see 3.3.11 and 3.4.5.4).

e. The levels of preservation, packaging, and packing applied to deliverable items shall be consistent with the known or projected method of delivery, program requirements, destination environment, logistics support plans, and shall be in consonance with any contractually required maintenance engineering analysis (MEA) or optimum repair level analysis (ORLA).

f. The performance of transportation and handling equipment and containers shall be continually evaluated during development testing to identify design and other configuration deficiencies, and corrective action taken as necessary to insure such equipment will perform in accordance with its approved baseline.

g. Protective packaging for dangerous and hazardous material shall be designed to meet the requirements of applicable directives.

3.1.1 Program scheduling. Program scheduling shall be accomplished, and packaging, handling and transportability management and technical events, and their projected time-phased accomplishment shall be depicted in milestone chart format in relation to overall program schedule. Slippages or other discrepancies between planned and actual performance, and reasons for changes in forecasts shall be reported to the procuring activity, supported by a

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brief narrative discussion of the cause(s). Items/equipment requiring SDPE development (see 3.3.11) shall be identified and tabulated and included as a part of this report. Updating the schedules shall occur as additional items of SDPE and transportability problem items are identified.

3.1.2 Work breakdown structure. The tasks required in complying with this specification and related statement-of-work provisions shall be identified in the work breakdown structure in accordance with MIL-STD-881.

3.2 Transportability. Transportability means the inherent capability of materiel (materiel, equipment, and item are used interchangeably throughout this specification) to be moved by towing, self-propulsion, or by carrier via railways, highways, waterways, pipelines, oceans, and air.

3.2.1 General design. Transportability shall be a primary consideration in the planning and designing of new or modified materiel to insure an efficient transportation capability by all required modes to achieve the required operational capability at the lowest program life cycle cost. Equipment designs which will require special or unique arrangement of schedules, rights-of-way, clearances, or other restrictive operating conditions shall be avoided insofar as possible (see 3.3). Handling, tiedown, and slinging points shall be incorporated in the design of equipment as needed to optimize transportability and complement maintenance, warehousing, and other handling requirements. Full design consideration also shall be given to the safety aspects of equipment and its sectionalization and disassembly capability for transport purposes, with ease of reassembly for operational use also maintained as a design objective. Also, self-propulsion shall be an equipment design consideration.

3.2.1.1 Mobility. Equipment intended to be mobile shall conform to MIL-M-8090 for the applicable type and class of mobility.

3.2.1.2 Standard equipment. Maximum practical use shall be made of standard handling/transportation equipment for compatibility with existing storage, transportation, handling, and maintenance facilities and equipment.

3.2.2 Identification and solution of transportability constraints. Design requirements that inhibit transportability shall be identified, and solutions proposed by the contractor during the initial system engineering processes utilizing the applicable procedures and documents of current configuration management directives and the appropriate data item description on the DD Form 1423. The process of transportability problem identification and solution shall begin in the planning/design phase, and shall continue throughout the complete program as necessary. Transportability problem items are described in 3.2.3. Table I lists numerous characteristics that affect transportability and will aid in identifying such problems.

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TABLE I. Item Characteristics Affecting Transportability

(Nonoperating or Inert State)

Physical Properties

Width	Net Weight
Height	Gross Weight
Length	Center-of-Gravity

Dynamic Limitations

Acceleration - Allowable acceleration, pulse time, and pulse shape along each of the mutually perpendicular axes.

Vibration - Critical resonant frequencies in plane of shipping attitude.

Deflection - Maximum allowable bending in plane of shipping attitude.

Skin Loading - Maximum allowable skin pressure loading diagrams.

Securing - Maximum allowable dynamic load on tiedown, mounting, and handling fittings along each of the mutually perpendicular axes.

Leakage - Maximum allowable solid, liquid, or gaseous emission rates.

Environmental Limitations

Temperature - Critical temperature range requiring controlled conditions.

Pressure - Critical pressure range requiring controlled conditions.

Humidity - Critical humidity range requiring controlled conditions.

Cleanliness and sterilization.

Hazardous Effects

Personnel Safety - Toxicity of fumes or liquids on contact with humans.

Radiation - Control of electromagnetic or radioactive radiation.

Electrostatic - Grounding requirements.

Explosives - Sensitivity to deterioration from impact or penetration.

Etiologic or Biologic - Personal or public health debilitation or lethality potential.

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3.2.3 Criteria. Design configurations which cause transportability problems are those which result in an oversize, overweight, fragile, dangerous, or hazardous item being offered for transportation. Transportability problem items are those items which:

a. Exceed either 8 feet in height, 8 feet in width, 32 feet in length, or 11,200 pounds in their transportation configuration or,

b. The materiel is to be transported in van or stake body trucks or cargo containers and exceeds any one or more of these conditions:

(1) Length - 18.5 feet

(2) Width - 6.5 feet

(3) Height - 7 feet

(4) Weight - 10,000 pounds.

c. Are sensitive and fragile to the extent that they require one or more of the following:

(1) Use of a single mode of delivery (or specialized equipment within modes) to protect them against damage during transit.

(2) Development or application of unique packaging or shock-mitigating devices or techniques for movement.

(3) Environmental control during transit or intransit storage to preclude damage or compromise of reliability.

(4) Monitoring by technical escorts during transit or intransit storage to preclude damage or compromise of reliability.

(5) Development or application of unique materials-handling devices or techniques specifically for the handling of the item.

(6) Furtherance of the state-of-the-art in transportation equipment design, materials-handling equipment design, or packaging technology before required delivery can be accomplished.

d. Are hazardous or dangerous to the extent that they require one or more of the following:

(1) A special permit from federal or state regulatory agency before movement can be made (ie, DOT, CAB, Bureau of Public Health, etc).

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(2) Monitoring by technical escorts during transit.

(3) Environmental control during transit or intransit storage to retain stability or preclude damage or compromise of reliability.

(4) Venting during transit because of the production of toxic or explosive fumes which cannot be contained by available packaging techniques.

(5) Movements of a rocket motor, class A or B explosive, in a propulsive state; reference title 49, Code of Federal Regulations, paragraphs 173.53(b) and 173.88(e).

e. The item is wheeled or tracked and is to be towed or self-propelled on or off highways.

3.3 General requirements, preparation for movement and handling. Requirements for peculiar configurations, routings, or exception(s) to the criteria listed herein that cannot be avoided in equipment design shall be brought to the attention of the procuring activity prior to final design release through updating the program schedule (see 3.1) or other appropriate data items when specified on the DD Form 1423.

3.3.1 Restraining systems. Items in their shipping configuration, including skidded or wheeled equipment, shall be provided with tiedown and lifting provisions commensurate with the size, weight, and mode of transportation. The carrier equipment involved shall also include similar features when necessary to insure safe and efficient handling and movement. Additional margins of safety shall be incorporated in the design of the restraining system when required because of peculiarities of the cargo, carrier safety considerations, or accident effects, especially where dangerous or hazardous materials are involved. Such provisions shall be appropriately depicted or described in related specifications and item design drawings. MIL-STD-209 establishes dimensional limits, design considerations, and positioning requirements for slinging eyes and attachments for lifting or tying down vehicles and other military equipment, shipped assembled or disassembled in unboxed or uncrated condition, and for tying cargo or accessories to transport equipment.

3.3.2 Air transportation. Items to be transported by air shall conform to the general design and performance requirements of MIL-A-8421 and, as far as practicable, shall be configured for compatibility with the 463L materials handling system (see 3.5). Cargo aircraft loading information is available in Bulletin 518 and appropriate technical orders in the -9 or -16 series for the applicable aircraft. (Contact procuring activity for handbook identification applicable to specific aircraft model/type when required.) Loading demonstrations shall be accomplished as required in accordance with MIL-A-8421.

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3.3.3 Highway transportation

3.3.3.1 Items in their shipping configuration to be transported via highways shall conform to the motor carriers handbook(s) of the states and the Limits of Motor Vehicle Size and Weights prepared by the International Road Federation for Foreign Countries through which the materiel will pass.

3.3.3.2 Vehicles intended for use on public highways shall comply with the established gross weight, axle, and tire load limits specified in federal and state vehicle safety statutes. The contractor's recommended routing shall consider statutory state and foreign highway limitations, seasonal limitations, underpass clearances, bridge loadings, load curve radii, etc.

3.3.4 Railway transportation. The average gross weight railroad car loadings range from 132,000 to 400,000 pounds. Loads in excess of this range may be transported when planned in terms of specialized cars, controlled speeds, and routing, which shall be coordinated with and approved by the procuring activity. For planning purposes, numbers and types of cars are listed in the Official Railway Equipment Register of the United States. Description and configuration of the cars may be found in the Car Builders Encyclopedia. Maximum dimensions and weight limitations for U. S. railways are contained in the Association of American Railroads Car Service Rule 14 of the current issue of the Railway Line Clearance publication of each individual railroad. Clearance dimensions and weight limitations applicable to railroads of foreign countries are contained in the Berne International Rail Interchange Agreement.

3.3.5 Water transportation. With the wide range of seaway and inland vessel configurations available, it is advisable to obtain information from commercial and military operators in the selection of specific water vessels. For planning purposes, consideration should be given to port facilities, seasonal pool levels, bridge clearances, shipping schedules, etc.

3.3.6 Induced environments

3.3.6.1 Carrier equipment. The increasing availability of military and commercial transportation vehicles and equipment with more sophisticated shock/vibration isolation systems, together with differing size, mass, and internal cushioning features has complicated the system designer's process for establishing standard and realistic guidance regarding shock and vibration environments of specific equipment. Where specific transportation vehicles are identified for use, their related shock and vibration environment shall be used.

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3.3.6.2 Hoisting, handling, and airdrop. Design consideration shall include factors encountered in:

3.3.6.2.1 Hoisting. Shock results from the application of lifting force. The shock extreme is considered to be 2g vertical for 11 to 40 milliseconds duration and may be incidental to any mode of transportation. Lifting rings or eyes shall be located to insure a stable lifting configuration.

3.3.6.2.2 Handling. Shock results from the handling, loading, and unloading operations and may occur prior to or subsequent to the use of any mode of operation. Unless otherwise specified, the frequency and type of handling to be encountered in world-wide distribution shall be a basic consideration in designing an adequate shock/vibration isolation system.

3.3.6.2.3 Airdrop. Shock results from the opening of the parachute and again from impact with the ground. If the ground delivery technique is employed, the initial shock will take place when the line attached to the aircraft cargo engages the ground anchor point. If materiel is to be prepared for airdrop, see MIL-STD-814 and MIL-STD-669. These standards establish detailed design, number and location requirements on airdrop materiel for tiedown to platform or aircraft, for suspension from retardation system, and for extraction from aircraft, loading environments, and related requirements.

3.3.7 Shock-vibration transmission. Packaging and transportation equipment used shall have the capability to insure that shock-vibration transmission is within the structural values established for the individual item or equipment.

3.3.8 Natural environment. Natural environment is produced by steady-state or recurring climatic conditions experienced in either transportation or storage, including sunshine, precipitation, air movement, radiation, magnetic gravitation, etc, which control variations in temperature, pressure, humidity, and movement or airborne solids. Climate extremes are contained in MIL-STD-210. However, since MIL-STD-210 represents the most severe conditions world-wide, if specific criteria is not included in the contract, system specification or contractually applicable system documents, the contractor shall be responsible for providing item protection based upon known transportation, handling, and storage conditions for deliverable items of the contract. Unless previously approved materials and processes are used, the contractor shall verify by demonstration, test, or engineering analysis that selected materials and protective processes provide item protection against the applicable climatic extremes involved.

3.3.9 Hazardous/dangerous materials. Protective packaging for dangerous and hazardous material shall be designed to meet the requirements of Titles 14, 46, and 49 of codes of Federal Regulations and AFM 71-4 as appropriate.

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Publications specified in ASPR 7-104.79 define safety precautions for ammunition and explosives and sets forth the directives which must be complied with in the preparation for delivery of such materials. Disassembly of an item to provide for separate packaging of the dangerous or hazardous portion shall be accomplished when effective and efficient in terms of system life cycle costs.

3.3.10 Uncrated shipments. Large items requiring special handling shall be shipped uncrated when they meet the criteria outlined in the appendix hereto. Items in this category not meeting these criteria shall be crated for protection and handling convenience.

3.3.11 Special design protective equipment (SDPE). SDPE is specialized equipment, itself a configuration item (CI), which is required for the support of another item when it is transported or stored during the manufacturing process, test phases, operation, and deployment. These are items normally identified during the system engineering process (MIL-STD-499), specification development (MIL-STD-490), or as a product of the transportability/packaging analysis. SDPE generally include trailers, certain dollies, skids, adapters, and engineered containers requiring formal design development and test. Equipment from the Department of Defense inventory requiring significant modification also shall be proposed by the contractor as SDPE.

3.3.11.1 Engineered containers. (For items of an especially complex nature, critically sensitive, hazardous, or fragile to the degree that a specialized container is required.) For purposes of this specification, a container shall be proposed as engineered (and documented as prescribed above for other SDPE) if it has special features such as combined handling capability, a unique energy damping system, controlled temperature, humidity or pressure functions, and which requires engineering drawings (MIL-D-1000) for the control of materials, dimensional tolerances, form, fabrication, and assembly. The design considerations specified herein and test requirements specified in section 4 shall apply.

3.3.11.2 Drawings and specifications. Specifications and drawings covering the SDPE (including engineered containers) shall be prepared in accordance with MIL-STD-490 and MIL-D-1000, respectively. Table II herein specifies supplemental data requirements for SDPE to be submitted with the CI development qualification for the equipment.

3.3.11.2.1 Configuration control. Submittal, review, and approval of the above and other packaging data referred to in 3.4.5.4 and 3.4.5.5 shall be in phase with the various reviews, audits, inspections, and baselines required by configuration management processes when configuration management is a statement-of-work requirement. Configuration control will be

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TABLE II. SDPE Design Approach Descriptions

Requirements Applicable to SDPE Design & Development Documentation (See 3.3.11) (This table also applicable to special study packaging (see 3.4.5.4)
For SDPE, a design evaluation drawing and supporting information described in b. below shall be included with the CI development specification for the equipment. The drawing shall include as a minimum the data set forth herein, as applicable. Component parts shall be identified by type, rating, etc. Cushioning material shall be described by specification, type, grade, thickness, and density.
a. The item shall be shown outlined in or on the proposed container or handling/shipping equivalent with the following data as it applies:
(1) Location and type of internal mounting or attaching provisions
(2) Vibration-shock isolation features, based on the predetermined fragility rating (or other constraint) of the item to be shipped - the fragility rating in g's of the item shall be included on the SDPE drawing
(3) Service items (indicators, relief valve, etc)
(4) Environmental control features (also specify method of preservation of MIL-P-116)
(5) External handling, stacking, and tiedown provisions with stress ratings
(6) Dimensional and weight data (gross and net)
(7) Bill-of-material
(8) Marking provisions, including the center-of-gravity location
(9) Position and travel of adjustable wheels, tilting, or other adjustments to facilitate loading
(10) For wheeled SDPE, give overall length, width, and height, with item mounted, turning radius, number of tires with pressure, and unit contact load.
b. Accompanying the SDPE development specification shall be:
(1) The results of trade studies (see 3.3.11.5)
(2) A detailed qualification test and demonstration plan (include proposed sequence of test methods and procedures)
(3) An indication of compliance with container standardization objective (see 3.4.5.1)
(4) An identification of the primary mode of transportation, or combinations of modes, and any peculiar or unusual transportation requirements in connection with the SDPE's design mission
(5) Significant interfacing transportation/materials handling equipment required for loading/off-loading/handling will be identified
(6) Outline of technical manual or handbook when required due to complexity of design. (Reference DI-M-3413/H-114.)

NOTE. The early initial submittal requirement for the above may preclude other than approximation in dimensional detail and related information; however, all follow-on data shall be definitive.

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exercised against established baselines in accordance with MIL-STD-480 or MIL-STD-481 as appropriate and as supplemented by MIL-STD-483. Additional guidance for the conduct of formal reviews and audits as applicable to the functional areas encompassed by this specification is provided in MIL-R-83313.

3.3.11.2.2 Interface. The CI specification number for an item of SDPE shall be referenced in the preparation for delivery section of the related CI specification which the SDPE will support.

3.3.11.2.3 Federal stock numbering. It is the intent that all SDPE shall be federally stock numbered, therefore, the contractor shall take stock numbering screening action in accordance with MIL-P-84000. The result of this action shall be furnished to the procuring activity at the time the SDPE is recommended for production procurement.

3.3.11.3 Design requirements. When it is determined that the SDPE is not available from the Department of Defense inventory (see 3.4.5.1), basic design requirements shall be established. Such requirements shall be established against, but not limited to, the transportability characteristics specified in table I, in correlation with the limiting factors and functional interfaces identified in the engineering documentation for the basic item it will support when such documentation is a contractual requirement. Other design considerations/requirements set forth herein are equally applicable.

3.3.11.4 Demonstrations. Insofar as practicable, the contractor shall interface the design and development of SDPE, and schedule reviews, inspections, and demonstrations to coincide with similar actions required for the related item which it will support. Demonstrations (see table II, item b.(2)) of marginal loading, tiedown, and movement conditions involving SDPE or other system/equipment items that are caused by close or critical tolerances shall be performed, and expedited when practicable, with dummy or mockup equipment.

3.3.11.5 Trade studies. Trade-off studies shall be accomplished and documented in accordance with MIL-STD-499 to the degree necessary to obtain the optimum container or other SDPE design in terms of program life cycle cost-effectiveness/performance. Studies shall identify significant features of the proposed SDPE that are subject to trade-off; eg, design configuration, environmental control features, transportability/handling provisions, safety features, and choice of materials selected such as wood vs metal vs plastic, ecological impact, etc. Contractors are encouraged to fully explore the latest techniques, processes, and state-of-the-art materials for use in container design, development, and fabrication. Consideration should be given to the long term value and reuse capability that may be provided by lightweight, high-stress, high impact materials that will retain structural integrity with minimum maintenance, particularly under extreme environmental conditions.

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3.4 Preservation, packaging, and packing (including container designs)

3.4.1 General considerations. In consideration of program life cycle costs, preservation methods, and packaging/container designs shall provide a reasonable balance between packaging cost and performance. The environmental control features, including preservation and packaging methods selected, shall insure protection of items against natural and induced environments. A prerequisite to such selection is the analysis of environments to which the item will be subjected during its life cycle. Such analysis shall include, but not be limited to, item characteristics, induced forces produced in transportation and handling, and climatic environments. System logistics, preoperational, operational, maintenance and material support plans provide basic guidance in performing these analyses.

3.4.2 General requirements

3.4.2.1 Preservation. Provisions for application of MIL-P-116 shall apply for selection of cleaning and preservation methods. Preservation selection charts contained in MIL-STD-794 shall be used in selecting appropriate methods.

3.4.2.2 Packaging and packing. Packaging and packing shall be of the appropriate level(s) or combination of levels required to satisfy the statement-of-work, or other procurement or approved planning document and shall be accomplished in accordance with MIL-P-116 and MIL-STD-794. The proper levels when not specified shall be selected from table III, subject to approval of the procuring activity. Insofar as possible and practical, blocking and bracing shall be in accordance with MIL-STD-1186.

3.4.2.2.1 Level C. While military specification materials, etc, are not normally required for level C, assistance in selecting adequate protective processes, materials, and containers for level C packaging and packing is available through the use of specifications listed in MIL-STD-794 and National Aerospace Standards in the NAS 850 series.

3.4.3 Marking for shipment. Packaging and container marking shall be in accordance with MIL-STD-129 and as stated herein. Special markings shall be as specified by the procuring activity.

3.4.3.1 Marking of dangerous materials. Marking and labeling of dangerous materials shall be in conformance to the pertinent applicable documents, procedures cited within this document, and general provisions of the contract. All interior packages and shipping containers for articles and materials classified as dangerous or restricted under provisions of Department of Transportation (DOT), Civil Aeronautics Board (CAB) or U. S. Coast Guard

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TABLE III. Selection of Levels of Protection for Storage or Shipment

	LIMITED TENURE OF STORAGE		FAVORABLE STORAGE		SHED OR OPEN STORAGE		UNKNOWN STORAGE		OVERSEA AIR SHIPMENT		OVERSEA WATER SHIPMENT		CONSOL'D or CONTAINERIZATION SHIPMENT		OVERSEA PARCEL POST	
	Pres-Pkg	Packing	Pres-Pkg	Packing	Pres-Pkg	Packing	Pres-Pkg	Packing	Pres-Pkg	Packing	Pres-Pkg	Packing	Pres-Pkg	Packing	Pres-Pkg	Packing
Shipments from contractors or other supply sources in CONUS to:																
a. CONUS requisitioners for immediate use w/no redistribution	C	C														
b. CONUS depots for storage and redistribution			See note 3										A	C		
(1) CONUS requisitioners			B/C	B/C	A	A	A	A								
(2) Oversea requisitioners			A/B	B	A	A	A	A								
c. CONUS contractors for assembly (CFP)	C	C	B/C	B/C	B	B										
d. Oversea requisitioners																
(1) In support of combat operations																
(a) For immediate use									A/B	A/B/C	A/B	A/B	A/B	A/B/C	A/B	B C
(b) Storage and redistribution anticipated			A/B	A/B	A	A	A	A	A/B	A/B	A	A	A/B	A/B	A/B	A/B
(2) Other than combat operations																
(a) For immediate use	B/C	B/C							B/C	B/C	A/B	A/B	A/B	B/C	B/C	C
(b) Storage and redistribution anticipated			B	B	A	A			See note 3 A/B	B	A/B	A/B	A/B	B	A/B	B

NOTES:

1. When a choice of levels is indicated, selection will be based on the following criteria:
   a. Level A
      (1) Ultimate destination is unknown; or
      (2) Duration or condition of storage is unknown or cannot be determined; or
      (3) Unfavorable transportation or handling conditions are known or anticipated; or
      (4) Open or shed type storage is known or anticipated; or
      (5) The item/shipment is known or anticipated to require the maximum degree of protection.
   b. Level B
      (1) Ocean shipments are intended for immediate use and favorable transportation, storage, and handling conditions are known to exist; or
      (2) Short term favorable storage and movement which do not involve ocean transportation; or
      (3) Storage tenure will not exceed the level B package shelf life when established by specifications.
      (4) Level B preservation-packaging shall not be used without approval of procuring activity.
   c. Level C
      (1) Movement, storage, and handling conditions are known to permit this level; or
      (2) Shipments moving via other than ocean transportation are for immediate use at the first receiving activity.
2. In selecting levels based upon transportation mode, consideration will be given to the additional modes of transportation and type of handling to which the item supplies may be subjected.
3. Shipments to humidity controlled storage facilities with no redistribution required - preserve, package, and pack level C.
4. Weapon System pouch shipments - domestic and overseas, when authorized by contract - preserve, package, and pack level C.
5. Requirements of this table are also applicable to overseas contractors.
6. Repackaging/repacking of items which have been preserved, packaged, or packed to a level which exceeds the requirement specified above is required when delay of shipment is permissible, and savings to be realized in packaging and transportation costs will exceed the total costs of such repackaging/repacking. All changes in preservation, packaging, and packing required in response to (amended or initial shipping instructions) shall be reported to the responsible administrative contracting officer for analysis. Any changes in cost of performance resulting from such instructions shall be processed in accordance with the clause of the applicable contract entitled "changes."

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Regulations, or AFM 71-4 shall be marked, regardless of exemption for mode of transportation, with proper shipping name of the item; flashpoint of all liquids having a flashpoint of 200°F or below; and the percentage of concentration of acids and corrosive liquids. In addition, the quantity of each dangerous or restricted material included in a container shall be annotated adjacent to the shipping name of the item, eg, acetic acid (80 percent concentration) - 1 quart. This includes the net weight of explosives contained in each interior and shipping container. Above informational marking shall be applied at the manufacturer's plant. When material is to be reshipped from a contractor's plant, the contractor shall be responsible for control to maintain identity during storage, handling, and reshipment.

3.4.4 Air packing. Air-type packing shall be in accordance with MIL-A-25175.

3.4.5 Design considerations

3.4.5.1 Container standardization. In selecting container designs, maximum effort shall be made to utilize containers or the container designs from those already available in the Department of Defense inventory, either in their existing configuration or with possible modification for new application, when this is the most cost-effective approach. Technical Information Centers and Defense Documentation Center (eg) are sources of information which will be utilized in determining available equipment and current state-of-the-art. Also, see federal stock catalogs.

3.4.5.2 Reusable containers. Reusable containers shall be proposed to the procuring activity by the contractor for items subject to repair, overhaul, or technical order compliance. The container design shall be in terms of the container retaining its usefulness with minimum maintenance through the projected life span of the item it supports. The design shall also provide for easy removal of the item from the container without degrading the containers total reusability value. Design data for these containers shall be prepared in accordance with 3.4.5.4 or if qualified as SDPE (3.3.11), then 3.3.11.2 applies.

3.4.5.3 Specification containers. Insofar as possible and practical, interior and exterior containers shall be selected from those referenced in MIL-STD-794.

3.4.5.4 Special study packaging. For items other than those supported by engineered containers or other SDPE that because of their physical characteristics require packaging that is sufficiently complex as to require drawings to describe, but which do not warrant the formal design, approval, control, and acceptance procedures required for SDPE, the following shall

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apply: Unless otherwise specified, data shall be prepared as Form 3 (MIL-D-1000) drawings in accordance with appropriate DD Form 1423 data item descriptions. The drawings shall include the technical characteristics specified in table II as they apply to the package design. The drawings shall be sufficiently detailed to permit effective evaluation of the proposed packaging, its duplication by a subsequent packager, and competitive reprocurement. The drawings shall be submitted to the procuring activity with the related CI specifications (when such specifications are applicable) with results of first article tests, or when acceptable to the procuring activity, statistical/analytical or experience information adequate to substantiate the capability of the proposed package to perform its intended function. Upon final approval, the drawings shall be reproduced in the preparation for delivery section of the CI product specification which the package supports.

NOTE: When specified in the contractual statement-of-work, the above procedures shall be applicable to level C packaging when it otherwise meets the criteria for special study packaging. In these instances, the capability of the package to provide adequate physical/mechanical protection to the item(s) shall be substantiated by rough handling tests in accordance with 4.2.1 herein - level B drop heights shall apply (FED-STD-101).

3.4.5.5 Section 5 requirements - CI specifications. The paragraph titled Preparation for delivery of MIL-STD-490 shall apply for all type A, B, or C CI specifications required by the contract or other procurement document. Detailed preparation for delivery requirements shall be included for each level of protection. Normally, levels A and C will constitute the levels for preservation and packaging. The provisions specified herein shall apply in preparing these requirements, including preparation and processing of packaging/container drawings. Reference by title or number only to the contractors internal processing or procedural documentation will not be acceptable. Specific portions of these internal processing or procedural documents which are pertinent to preparation for delivery instructions for an item must be defined and included as an integral part of the section 5. This is equally applicable to data prepared for SDPE (3.3.11).

3.4.5.6 Performance verification. Selected containers, packages, and handling/shipping devices and equipment shall be capable of retaining established end item reliability and structural integrity. The contractor shall conduct the applicable tests of section 4 of the specification and assure that the container or other device/equipment is capable of performing its design mission when subjected to these tests.

3.4.5.7 Protection of the contained item. The container or other shipping/handling device and its suspension or cushioning media shall protect the

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item from dynamic environments induced during shipping and handling. Movement of the item within the container shall be controlled to prevent contact with the container shell or support structure.

3.4.5.8 Determination of item fragility. Dynamic values in the nonoperating state established in end item specifications shall be used to establish the maximum energy (normally expressed in g's) that can be permitted to reach the item during transportation and handling. It shall be necessary to provide a means of absorbing induced energy so as to limit shock values to or below the specification limits. Care shall be taken to insure that there are no additional components installed with a lower fragility rating or that will alter the structural strength characteristics downward from conditions underwhich the prime item was tested and the shipping fragility was established.

3.4.5.9 Energy damping methods and package cushioning design. Design and application of conventional cushioning systems and selection of the appropriate cushioning material shall be in accordance with MIL-HDBK-304.

3.4.5.9.1 Resilient cushioning. Material used for cushioning applications in the form of rolls, sheets, special-cut pads, or molded forms shall comply with MIL-C-26861.

3.4.5.9.2 Resilient mounts. Resilient mounts used for suspending such items as engines, transmission, navigational or communications equipments, and similar items from their provided mounting or installation points shall comply with MIL-M-45907.

3.4.5.9.3 Equipment mounts. Equipment with integral vibration-shock mounts designed for use in airborne and ground equipment shall not be shipped on the active shock mounts. The mounts shall be made inactive and a suitable cushioning system provided.

3.4.5.10 Container accessories. Valves (except drain valves), humidity indicators, record receptacles, etc, that are subject to routine inspection shall be grouped in one end of the container, unless the size of the container would warrant placement of an accessory item in more than one location. All covers for such features shall be captive.

3.4.5.11 Relief valves. Vacuum relief, pressure relief, or vacuum-pressure relief valves shall conform to MIL-V-27166.

3.4.5.12 Drain valve(s). Drain valve(s) shall be located as required in the container to provide drainage of fluids, moisture, etc.

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3.4.5.13 Humidity indicators. Humidity indicators required for mounting in rigid containers of flexible barriers shall conform to MIL-I-26860, and shall be located to permit unrestricted container airflow to the indicator for accuracy of the indication. Card-type indicators placed within a barrier for viewing through provided inspection windows shall conform to MS20003.

3.4.5.14 Record receptacle. When a containerized item must be accompanied by records/log/books/manuals, etc, an exterior record receptacle shall be provided. Cylindrical receptacles shall have minimum inside dimensions of 2 inches diameter by 9 inches long. Rectangular receptacles shall have minimum inside dimensions of 2 inches deep, 9 inches wide, and 12 inches high. Closure shall be watertight. Drain holes shall be provided as required to permit drainage of accumulated moisture.

3.4.5.15 Desiccant. Desiccant shall be so placed in the container that it will be readily accessible for removal when redesiccation is necessary. When feasible, separate entry(s) shall be incorporated in the container to eliminate the requirement for opening the main container assembly solely for this purpose. Desiccant shall not be placed in packages or containers so as to support any load.

3.4.5.16 Components. Accessory items, handling and fastening devices, tiedown and hoisting fittings, and other integral components of the container when not in use should not extend beyond the basic profile or envelope of the equipment, unless shields are provided.

3.4.5.17 Opening and closure. Container closures shall be designed to permit rapid opening and closing without the use of special tools by unskilled personnel, and shall be easily operable in the climatic environment in which the container is intended to function. Reusable closure fasteners shall be a design consideration.

3.4.5.18 Stacking capability. The design of containers intended to have an inherent stacking capability shall be such that it will insure a stable stacking configuration. Stacking stability provisions shall include positive means for discouraging relative displacement under impact loads encountered in shipment or multiple unit handling.

3.4.5.19 Safety. Container designs (or other shipping/storage device) for hazardous or dangerous materials shall be of the highest standards of safety so that failure of the containers/device will not create a hazardous condition by reason of its mode of failure. Safety capability shall be demonstrated to the satisfaction of the government.

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3.4.6 Protection of rugged equipment. Wheeled equipment, items designated for use in the open, and other rugged equipment which are acceptable to the carrier in an as is condition shall not require crating. When corrosion protection is required, preservation shall comply with MIL-P-116.

3.4.7 Value engineering. The cost of packaging methods, procedures, and SDPE shall be continuously evaluated during the design stage to obtain an optimum benefit/life cycle/cost ratio. Evaluations shall be conducted in accordance with MIL-V-38352 or ASPR 1-170.1 as contractually applicable and shall continue through operational test and evaluation (OT&E) and production.

3.4.8 Reliability. The contractor shall insure that selection of preservation and packaging methods and container designs are capable of maintaining the established reliability levels of the items being packaged. Retention of the reliability of the item during storage and transportation shall be major considerations in accordance with MIL-STD-785, and its provisions shall apply through OT&E testing and production.

3.4.9 Ecological impacts. Selection and use of packaging materials and techniques shall consider the impact of such selection on the consumption of natural resources, waste, package/material disposal, pollutants expelled into the atmosphere and any other potentially detrimental ecological impacts.

3.4.10 Field force protection. All items, components, and assemblies (eg, transistors, diodes, resistors, microcircuits, etc) which are subject to degradation by field forces (electrostatic, electromagnetic, magnetic, or radioactive) shall be provided protection as specified in MIL-STD-794 or the appropriate process or commodity specification (eg, MIL-S-19491 and MIL-M-55565).

3.4.11 Dangerous material. The preparation for delivery of dangerous materials, including special procedure for securing approval for shipment of rocket motors (class A and B explosives) in the propulsive state, shall be in accordance with the general provisions of the contract and as further stated herein.

3.4.11.1 Items subject to ignition detonation by electrostatic discharge. All items which are subject to ignition/detonation by electrostatic discharge and which are to be packaged in MIL-B-121, MIL-B-131, MIL-F-22191 materials, or other nonconductive plastic or plastic-coated materials first shall be individually wrapped in conductive material. A DD Form 1387-2 (MIL-STD-129) bearing the following notation shall be affixed to each unit package:

> WARNING (red letters). Contents subject to ignition or detonation by electrostatic discharge. GROUND inner conductive enclosure before and during removal from this package.

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Shrouds or covers protecting large items such as engines which are subject to ignition/detonation by electrostatic discharge, shall be of conductive material and shall be grounded during installation and removal of the shroud or cover and during transportation, receiving, and storage. A DD Form 1387-2 bearing the following notation shall be affixed to each side of the shroud or cover:

> WARNING (red letters). This item subject to ignition or detonation by electrostatic discharge. GROUND this cover/shroud during installation and removal and during transportation, receiving, and storage.

Contractors required to perform within the purview of the above are authorized to use material conforming to MIL-P-81705.

3.5 463L materials handling support system (see 3.3.2)

3.5.1 Preparation for military air transportation. When air transportation is identified or contemplated for system test, deployment, operation or support, the contractor shall assure proper interface between equipment designs and the 463L system. This applies to nontactical as well as tactical systems, since military cargo aircraft utilization rates are dependent on the improved turn-around-time made available with roller-pallet equipment. Palletizing of smaller packaged or unpackaged equipments is a prime requisite for logistical, operational, and deployment operations. Loading/off-loading and unit load distribution over the aircraft roller system is enhanced when large equipments are provided with a suitable load conversion device (ie, skids, wheels, pallets, etc).

3.5.2 System description. The 463L materials handling support system consists of five separate, but interdependent equipment families. These are air cargo terminals, cargo preparation, cargo ground handling, aircraft cargo loading, and intransit control facilities. Each family is equipped with roller conveyors permitting movement of cargo between the various segments of each family without the use of additional handling aids. Roller conveyors in cargo aircraft and ground handling equipment are designed to accept standard-size pallets.

3.5.3 Pallets and nets. Characteristics and identification of pallets and nets are as follows:

a. HCU-6/E aircraft cargo pallet

(1) Conforms to MIL-P-27443, type I

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(2) Dimensions: 108 by 88 by 2-1/4 inches

(3) Rated capacity: 10,000 pounds

(4) Weight: Not to exceed 298 pounds

b. Net assembly for HCU-6/E pallet - The net assembly consists of an HCU-15/E top net and two HCU-7/E side nets conforming to MIL-N-27444.

(1) The top net weighs 21 pounds, and each side net weighs 22 pounds

(2) Hardware fittings are provided around the base of the side net for attachment to the pallet tiedown rings and around the top of the side net for attachment of the top net

(3) The maximum rated capacity of the assembly is 10,000 pounds

c. HCU-10/C aircraft cargo pallet

(1) Conforms to MIL-P-27443, type III

(2) Dimensions: 54 by 88 by 4-1/2 inches

(3) Rated capacity: 5,000 pounds

(4) Weight: Not to exceed 215 pounds

d. Net assembly for HCU-11/C and HCU-12/E pallets - The net assembly consists of an HCU-16/C top net and two HCU-11/C side nets conforming to MIL-N-27444.

(1) The top net weighs 22 pounds, and each side net weighs 14 pounds

(2) Hardware fittings are provided around the base of the side net for attachment to the pallet tiedown rings and around the top of the side net for attachment of the top net

(3) The maximum rated capacity of the assembly is 5,000 pounds

e. HCU-12/E aircraft cargo pallet

(1) Conforms to MIL-P-27443, type II

(2) Dimensions: 54 by 88 by 2-1/4 inches

(3) Rated capacity: 5,000 pounds

(4) Weight: Not to exceed 160 pounds.

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3.5.4 Aircraft utilization. Types C-141, C-135, C-130, and C-5 cargo aircraft are either equipped with or are compatible with the 463L system. Consult Bulletin 518 for cargo clearance in these aircraft. Head clearance is affected by the 463L system.

3.6 Materiel support

3.6.1 General. Preservation, packaging, packing, and marking for the materiel support process during phase testing and production shall be of the appropriate levels to satisfy storage and shipping requirements at the contractor's storage site, test site, and during the production phase of system acquisition.

3.6.2 Packaging requirements for operational test and evaluation

3.6.2.1 Contractor-furnished equipment. Serviceable materiel delivered to the contractor's storage site or Air Force test site from the prime contractor or his subcontractors or vendors shall be provided sufficient protection commensurate with anticipated or actual environments to insure retention of the item or equipment reliability throughout testing phase(s). Table III provides guidance. The preparation for delivery requirements of the appropriate level as described in the approved applicable CI specification shall be used for this purpose when available.

3.6.2.2 Government-furnished equipment. Materiel received by the prime contractor from government suppliers or the Department of Defense inventory that has been preserved and packaged in accordance with level C shall be repackaged by the prime contractor to level A when extended storage periods or other conditions dictate the need for increased protection to insure that serviceability of materiel is maintained. Packing requirements shall be commensurate with transportation mode, storage, tenure, and other destination considerations.

3.6.2.3 Inspection of assets. In the event items in the contractor's storage facility are unpackaged or unpacked for inspection purposes, the preservation and packaging shall be restored to the level commensurate with the above requirements.

3.6.2.4 Excess and residual material

3.6.2.4.1 Excess and residual material. Serviceable or technical order compliance (TOC) excess or residual parts or equipment returned to the government shall be preserved, packaged, and packed in accordance with original item packaging data unless otherwise specified by the government. Where no data is provided, packaging shall be level A (see preservation selection chart, MIL-STD-794). Packing shall be level A where the unit container is the shipping container, otherwise level B packing shall apply.

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3.6.2.4.2 Reparable excess or residual parts/equipment. Reparable excess or residual parts/equipment for which packaging has not been specified by the government shall be provided level C protection as required to prevent further deterioration due to rust, corrosion, or physical damage. Spare reparable assemblies or components replaceable as a unit shall be individually packaged and provided similar protection with tamperproof seals provided when required.

3.6.3 Packaging requirements for operational test and evaluation and production

3.6.3.1 General requirements

3.6.3.1.1 Packaging and packing. Items programmed for delivery to the Air Force inventory or using commands shall be preserved, packaged, and packed at the level or combinations of levels necessary to insure the retention of items/equipment inherent reliability through the projected transportation and storage environments. The levels selected shall be commensurate with program/contract requirements. Section 5 of approved CI specifications (MIL-STD-490) or other government approved packaging data/specifications shall be used for this purpose unless otherwise stipulated by the procuring activity.

3.6.3.1.2 Packaging of kits. Modification and parts kits shall be packaged in accordance with MIL-STD-794.

3.6.3.1.3 Quantity unit pack. Unless otherwise specified by the procuring activity, the quantity of spare parts contained in a unit package shall be determined from instructions specified in MIL-STD-794.

3.6.3.2 Packaging data. Preservation and packaging data for items that are procured for the operational inventory shall be prepared and submitted for approval as contractually committed on a DD Form 1423. When transportation packaging orders (TPOs) are required and packaging drawings for the same items have been developed and approved in accordance with 3.4.5.4, the drawings shall be submitted in lieu of the TPO required by MIL-STD-834. This does not negate the requirement for submission of the related figure 1 format or PCAM card as applicable. The prime contractor shall insure that all packaging data submitted (including that from lower-tier contractors) as the result of various contract data items are compatible, prior to those data being submitted to Air Force agencies. For example, packaging data submitted in accordance with MIL-STD-834 shall be consistent with the packaging requirements specified in CI specifications for the same item.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Formal qualification testing. Unless otherwise specified by the CI specification or other contract requirement, formal qualification testing shall be conducted in accordance with MIL-STD-483 on one package, container, or SDPE representative of those to be supplied under the contract to demonstrate its capability to perform its intended function. Inspections shall be conducted at the contractor's plant under the responsibility of the contractor and in accordance with the test procedures specified herein. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the procuring activity. The contractor shall inform the procuring activity when tests are to be conducted so that a representative may be designated to witness and monitor the tests when so desired. First article inspection of new packaging methods is required when the method is a departure from previously developed preservation and packaging having a performance record for similar or identical items.

4.2.1 Rough handling tests. Rough handling tests for the appropriate level shall be accomplished in accordance with MIL-STD-794 utilizing the test methods, criteria, and related procedures prescribed therein. Also see note in 3.4.5.4.

4.2.1.1 Environmental conditioning. Equipment (containers, packaging materials, etc) likely to be affected by temperature extremes shall be conditioned prior to testing at those temperatures peculiar to their projected environment.

4.2.1.2 Vibration test. Selection of the proper vibration test shall be determined by the conditions a particular package or container is considered most likely to encounter (see FED-STD-101, Methods 5019 and 5020). When vibration testing is required, it shall be accomplished prior to conducting other rough handling tests.

4.2.2 Other tests. Other tests shall be performed as required to verify the adequacy of the package/container to accomplish its intended mission (see 3.3.8).

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4.2.3 Thermal tests. When thermal protection of an item is required, its container shall be tested to determine insulating properties in addition to test requirements of 4.2.1 and 4.2.2. Temperature values and ranges may be adjusted from specification requirements to coincide with temperature sensitivity established for the item.

4.2.4 Pressure control. When containers are required to provide a pressure control for the item, the container shall be tested to determine pressure control capability.

4.2.5 Testing of mobile SDPE. Mobile SDPE shall be tested to verify that induced forces reaching the item correspond to the limits established in the end item detail specification. Both the item and vehicle shall be instrumented with laboratory quality equipment having an accuracy which conforms to the calibration standards established by MIL-C-45662. Additional test requirements may be imposed by the detail specification or the approved test plan (see table II, item b(2)).

4.2.5.1 If the CI is itself a vehicle, it shall be tested in accordance with MIL-M-8090 for the type and class of mobility corresponding to its anticipated service use.

4.2.5.2 If the item is equipment designed to be mounted on or transported by a vehicle, it shall be mounted or loaded on a test vehicle and tested as follows. The test vehicle and the tests shall be selected for the most severe MIL-M-8090 specified type and class mobility consistent with the anticipated service of the vehicle on which the equipment is designed to be mounted or transported. The road types and terrain conditions specified for testing shall be as defined in MIL-M-8090.

4.2.6 Test reports. The results of first article and quality conformance inspections shall be prepared in accordance with MIL-STD-831 and distributed as specified by the procuring activity on the DD Form 1423.

4.2.7 Functional test. Functional tests shall be conducted for all end items tested in accordance with 4.2.1, 4.2.1.2, 4.2.2, 4.2.3, 4.2.5, and 4.2.5.2.

4.3 Quality conformance inspection (QCI). QCI/tests shall consist of all the tests deemed necessary to determine that the packages or containers submitted for acceptance under the contract are equivalent in performance and construction to the approved FAI items. In the absence of packaging process specifications, testing and sampling, except for SDPE, shall be in accordance with MIL-P-116. Rough handling tests shall be accomplished in accordance with MIL-STD-794.

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5. PREPARATION FOR DELIVERY

5.1 This section is not applicable to this specification.

6. NOTES

6.1 Intended use. This specification is intended to provide requirements to contractors in identifying and resolving transportability and packaging problems resulting from constraints imposed on systems, system segments, and related items/equipment. It also provides other general requirements in preparing material, including dangerous/hazardous, for shipment and storage as required for development testing and the production program.

6.2 Miscellaneous. AFSC DH 1-2, General Design Factors, Chapter 5, provides additional guidance in preparing materiel for shipment and storage. It also describes factors that should be considered in the item design process that would assist in developing economical and simplified package designs and enhance transportability.

6.3 Data delivery. Data generated as a result of this military specification being placed on contract is not deliverable unless specified on the contract data requirements list (DD Form 1423) referencing the appropriate data item description (DID) in TD-3. The data recommended to be placed on the CDRL when utilizing this specification is as follows:

Paragraph No.	DID
3.1.1, 3.3	DI-A-3007/M-103 or DI-A-3009/M-105 as appropriate
3.2.2, 3.3	DI-L-3327/P-104 and DI-S-3604/S-126 through DI-S-3608/S-130 as appropriate
3.3.11.2	DI-E-3102/C-102 and DI-E-3103/C-103
3.4.5.2, 3.4.5.4	DI-E-3100 series/E-100 series as appropriate
Table II	DI-T-3714/T-115 and DI-M-3413/H-114
3.3.11.2.1	DI-E-3128/C-141
3.3.11.5	DI-S-3606/S-128
3.6.3.2	DI-L-3305/L-107
4.2.6	DI-T-3721/T-125.

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6.4 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewers:
Air Force - 10, 13, 19, 69

Preparing activity:
Air Force - 11

Project No. PACK-F020

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APPENDIX

UNCRATED SHIPMENT OF AIR FORCE PROPERTY REQUIRING SPECIAL HANDLING

10. SCOPE

10.1 The purpose of this appendix is to provide guidance for determining whether large items of Air Force materiel requiring special handling should be shipped crated or uncrated.

10.1.1 A qualified carrier as used herein is a carrier certified to transport items which are not crated. Lists of qualified carriers may be obtained from Defense Traffic Management Services Regional Offices.

20. APPLICABLE DOCUMENT

20.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this appendix to the extent specified herein:

REGULATION

Department of Defense

4500.32-R Military Standard, Transportation and Movement Procedures

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 Large items requiring special handling shall be shipped uncrated when they meet the following criteria. Items not meeting these criteria must be crated for protection and to facilitate handling.

a. The cost of packing and shipping by other means would exceed the cost of shipment in an uncrated manner, or the time saved justifies this method of shipment.

b. The item(s) will withstand shipment in an uncrated condition and qualified carriers are available to handle the shipment.

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c. The destination is equipped to accept uncrated shipments.

d. The security classification of the item does not require packing or crating for concealment to preclude disclosure or inspection by unauthorized personnel.

e. The item is not classified as dangerous or hazardous cargo, subject to packing in accordance with Department of Transportation or other regulatory agency regulations.

30.2 Parts or components which are highly susceptible to damage or pilferage shall be removed from the major component and adequately packaged and marked to protect against damage, theft, or loss. These parts or components shall be securely attached to the unit being shipped prior to the release to the carrier.

30.3 Other items may be loaded in a vehicle which is partially filled with uncrated items, provided they are properly packaged, braced, and blocked.

30.4 DD Forms 1387-2 must be prepared to accompany uncrated shipments. These forms must be annotated to indicate that shipment is in compliance with the provisions of this specification. (See DOD Regulation 4500.32R.)

30.5 If reshipment of uncrated articles becomes necessary, packaging instructions must be obtained from the AFSC procuring activity or the Air Materiel Area inventory manager having responsibility for this items involved.

40. RESPONSIBILITY

40.1 The contractor when performing as the shipping activity:

a. Must make cost comparisons in compliance with 30.1a.

b. Must determine the capability of destinations to receive proposed uncrated shipments. Receiving activities must have suitable unloading equipment, storage facilities, and a capability for reassembly of the item when required.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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